MILITARY SPECIFICATION

RADIO SET AN/URC-80(V)5

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification defines minimum performance requirements for the AN/URC-80(V)5, a multichannel radio communications set (transceiver) utilizing frequency modulation in the 156-162 megahertz (MHz) frequency range. The AN/URC-80(V)5 shall provide voice communications subject to the FCC rules and regulations associated with Shipboard Stations in the Maritime service.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

MILITARY

MIL-E-17555 - Electronic and Electrical Equipment Accessories and Repair Parts, Packaging and Packing of

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes

MIL-STD-109 - Quality Assurance Terms and Definitions

MIL-STD-167 - Mechanical Vibrations of Shiboard Equipment

MIL-STD-471 - Maintainability Demonstration

MIL-STD-781 - Reliability Test: Exponential Distribution

MIL-STD-794 - Parts and Equipment, Procedures for Packaging and Packing of

MIL-STD-1472- Human Engineering Design Criteria for Military Systems, Equipment and Facilities

MIL-STD-1600- Reliability Prediction of Monolithic Integrated Circuits

HANDBOOKS

MILITARY

MIL-HDBK-217 - Reliability Stress and Failure Rate Data for Electronic Equipment

MIL-HDBK-472 - Maintainability Prediction

PUBLICATIONS

MILITARY

NAVPERS 15105 - Manual of Navy Enlisted Classifications

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

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2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

ELECTRONIC INDUSTRIES ASSOCIATION

RS-152 (Feb 1970) - Minimum Standards for Land-Mobile Communications FM or PM Transmitter, 25-470 mc.

RS-204 (July 1972) - Minimum Standards for Land-Mobile Communications FM or PM Receivers

(Application for copies should be addressed to the Electronic Industries Association, Engineering Department, 2001 Eye Street, N.W., Washington, D.C. 20006).

FEDERAL COMMUNICATIONS COMMISSION

FCC Rules Part 83

(Application for copies should be addressed to the Superintendent of Documents, U.S. Government Printing Office, Washington, D.C. 20402).

3. REQUIREMENTS

3.1 General. The AN/URC-80(V)5 shall be form, fit, and functional with the government furnished equipment (GFE). (see 6.3)

3.1.1 Terms and definitions. Terms and definitions shall be in accordance with RS-204 and RS-152.

3.1.2 Description. The AN/URC-80(V)5 shall contain a receiver/transmitter capable of providing 55 selectable transmitting channels and 61 selectable receiving channels. The radio set shall be designed for both simplex operation (f1/f1) (see 6.3) and half duplex operation (f1/f2) (see 6.3) in accordance with the channelization requirements of 3.1.8 over the frequency range of 156.025 MHz to 162.550 MHz. In addition to a selectable receiver/transmitter section, the URC-80(V)5 shall also have one operated guard receiver fixed tuned to 156.800 MHz. The URC-80(V)5 shall operate from a single antenna with both receivers operating continuously except during transmissions.

3.1.3 Operational control. The R/T unit shall be capable of being remotely controlled via cable from a distance of at least 25 feet (762.00 cm) by a control unit which performs the functions provided by the controls and switches of 3.2.5.1.

3.1.4 Equipment. The radio set shall be complete, ready for installation and operation, and shall include installation brackets and mating connectors capable of handling RG8 cabling from the antenna. The transceiver shall have as a minimum the following basic items:

1. Receiver/Transceiver unit RT-1165 (including)
   a. Receivers (2)
   b. Transmitter (1)
   c. Power Supply
2. Control unit (C8980/URC-80(V))
3. Loudspeaker (LS609/URC-80)
4. Antenna (AS3015/UR)
5. 25 feet (762.00 cm) cable (control unit to R/T unit)
6. Handset H-169/u

3.1.5 FCC rules. The AN/URC-80(V)5 shall have been type accepted and certified under the FCC rules and regulations governing stations on shipboard in the Maritime service (Part 83), as ammended, for operation after 1 January 1973.

3.1.6 System deviation and channel spacing. The AN/URC-80(V)5 shall be designed for narrowband operation. Narrowband operation shall be operation with RF channels spaced 25 kilohertz (kHz) apart with a maximum frequency deviation of ±5 kHz at 100 percent modulation.

3.1.7 Emission characteristics. Emission characteristics shall be 16F3 for narrowband operation in accordance with FCC Rules, Part 83.

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3.1.8 Channelization. The guard receiver shall be single channel, fixed tuned, crystal controlled and aligned to operate on 156.800 MHz prior to shipment. The main transmitter and receiver shall be aligned to operate on the following frequencies. Radio set shall be capable of operating in both the U.S. and International modes of operation. The transmitter section shall be inhibited on the channels indicated in Table I.

TABLE I - NARROWBAND

CHANNEL	TRANSMITTING FREQUENCY (MHz)	RECEIVING FREQUENCY (MHz)	
		US MODE	INT'L MODE
00	Inhibit	160.600	
01	156.050	160.650	
02	156.100	160.700	
03	156.150	160.750	
04	156.200	160.800	
05	156.250	160.850	
06	156.300	156.300	
07	156.350	156.350	160.950
08	156.400	156.400	
09	156.450	156.450	
10	156.500	156.500	
11	156.550	156.550	
12	156.600	156.600	
13	156.650	156.650	
14	156.700	156.700	
15	156.750	156.750	
16	156.800	156.800	
17	156.850	156.850	
18	156.900	156.900	161.500
19	156.950	156.950	161.550
20	157.000	161.600	
21	157.050	157.050	161.650
22	157.100	157.100	161.700
23	157.150	157.150	161.750
24	157.200	161.800	
25	157.250	161.850	
26	157.300	161.900	
27	157.350	161.950	
28	157.400	162.000	
29	Inhibit	157.450	162.050
60	156.025	160.625	
61	156.075	160.675	
62	156.125	160.725	
63	156.175	160.775	
64	156.225	160.825	
65	156.275	156.275	160.875
66	156.325	156.325	160.925
67	156.375	156.375	
68	156.425	156.425	
69	156.475	156.475	
70	156.525	156.525	
71	156.575	156.575	
72	156.625	156.625	
73	156.675	156.675	
74	156.725	156.725	
75	Inhibit	156.775	
76	Inhibit	156.825	
77	156.875	156.875	
78	156.925	156.925	161.525
79	156.975	156.975	161.575
80	157.025	157.025	161.625
81	157.075	157.075	161.675
82	157.125	157.125	161.725
83	157.175	157.175	161.775
84	157.225	161.825	
85	157.275	161.875	
86	157.325	161.925	
87	157.375	161.975	
88	[illegible]	[illegible]	
89	Inhibit	[illegible]	
[illegible]	[illegible]	[illegible]	

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3.1.9 Cabinet. The AN/URC-80(V)5 shall be designed for shelf mounting without vibration or shock mounts.

3.1.10 Solid state design. All active circuits in the transceiver shall be solid state.

3.1.11 Frequency synthesis. Generation of all channels shall be accomplished by frequency synthesis. Changing of channels shall not require the removal or insertion of crystals, other components, or realignment.

3.1.12 Antenna overload. The transmitter final amplifier shall not be damaged under variable load conditions such as grounding, partial grounding, or disconnecting of the antenna. An input signal (10 VRMS at any frequency at the antenna jack) shall not damage the receiver.

3.2 Electrical requirements.

3.2.1 Power source. The AN/URC-80(V)5 shall be designed to operate from a 117 Vac, 60 Hz ± 3 Hz power source. Equipment performance shall meet the requirements of the Power supply voltage range minimum standard paragraph of RS 152 for transmitters, and RS204 for receivers.

3.2.1.1 AC power cable. No AC power cable shall be supplied but AC mating connector to R/T unit is required.

3.2.2 Transmitter performance requirements. The transmitter shall meet all requirements herein under the Standard test conditions paragraph of EIA RS-152 for narrowband operation on all RF channels throughout the channelization specified. (see 3.1.8).

3.2.2.1 Rated RF power output. The transmitter shall meet the RF power restrictions listed in the FCC Rules. Power output shall be constant between 20 to 25 watts (w) reducible to 0.75w to 1.0w by means of a power level switch. The transmitter shall be designed for alignment and operation with load impedances corresponding to VSWR's from 1.0:1 to a maximum of 3:1 with respect to 50 ohms. The transmitter shall have an output impedence of 50 ohms. Carrier power shall not drop below maximum carrier power output during a duty cycle of one minute "ON" and one minute "OFF" for a period of eight hours.

3.2.2.2 Conducted and radiated spurious emissions. The conducted and radiated spurious emissions shall be greater than 90 dB below rated RF power.

3.2.2.3 Audio frequency harmonic distortion. The audio frequency harmonic distortion shall not exceed 5 percent.

3.2.2.4 Audio frequency response. The audio frequency response shall be in accordance with RS-152.

3.2.2.5 FM hum and noise level. The FM hum and noise level shall be at least 40 dB below standard test modulation.

3.2.2.6 Modulation limiting. The modulation limiting shall be in accordance with RS-152.

3.2.2.7 Carrier frequency stability. The carrier frequency stability shall be within ± .0005 percent of the assigned carrier frequency over a minimum 10 hour period when tested to the limits specified in 3.2.1, 3.2.2.8, and 3.2.4.

3.2.2.8 High humidity. The requirement limit shall be in accordance with RS-152.

3.2.2.9 AM hum and noise. The requirement limit shall be in accordance with RS-152.

3.2.2.10 Transmitter sideband spectrum. The requirement limit shall be in accordance with RS-152.

3.2.3 Receiver performance requirements. Simultaneous operation from the radio set antenna of the two receivers shall not cause degradation in performance of either receiver below the performance specified herein. The input impedance of each receiver shall be 50 ohms. Minimum standards listed in RS-204 shall be disregarded, unless otherwise [illegible] the receiver meeting the requirements specified herein. The receivers shall meet the requirements for narrowband operation in accordance with the channelization requirements in 3.1.8.

3.2.3.1 Usable sensitivity. The usable sensitivity shall be 1.0 microvolt (open circuit) maximum.

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3.2.3.2 Quieting sensitivity. The quieting sensitivity shall be 1.0 microvolt (open circuit) maximum.

3.2.3.3 Audio squelch sensitivity. The threshold sensitivity shall be 0.75 microvolt (open circuit) maximum.

3.2.3.4 Modulation acceptance bandwidth. The modulation acceptance bandwidth shall be greater than ± 7 kHz.

3.2.3.5 Adjacent channel selectivity and desensitization. The adjacent channel sensitivity and desensitization shall be 70 dB minimum.

3.2.3.6 Spurious response attenuation. Spurious response attenuation shall be a minimum of 100 dB.

3.2.3.7 Intermodulation spurious attenuation. The requirement limit shall be as specified in the FCC Rules, part 83, as amended.

3.2.3.8 Audio power output. The maximum audio power output shall be a minimum of 5.0 watts at no more than 5 percent distortion.

3.2.3.9 Audio frequency response. The audio frequency response shall be in accordance with RS-204.

3.2.3.10 Hum and noise. The hum and noise ratio shall be in accordance with RS-204.

3.2.3.11 Undesired conducted power. The undesired conducted power shall be in accordance with RS-204.

3.2.3.12 Undesired radiated power. The undesired radiated power shall be in accordance with RS-204.

3.2.3.13 High humidity. The amount of receiver degradation shall be no greater than that specified in 3.2.4 when the receiver is tested in accordance with the High humidity test paragraph of RS-204.

3.2.3.14 Frequency stability. Frequency stability shall be within ± 0.001 percent of the assigned frequency over a minimum of 10 hours when tested to the limits specified in 3.2.1, 3.2.3.13, and 3.2.4.

3.2.4 Temperature range. The equipment shall be operable over an ambient temperature range of 0°C to + 50°C with the maximum degradations listed in Table II.

3.2.5 Functional control performance requirements.

3.2.5.1 Controls. The control unit shall have as a minimum the following functional switches and controls:

a. Speaker volume level control (one for each receiver)
b. Squelch sensitivity control (one for each receiver)
c. Channel selector
d. High/low RF power switch
e. AC power ON/OFF switch (AC power switch may be located on R/T unit)
f. Mode switch (U.S./International)
g. Dimmer switch

3.2.5.2 Indicators. The control unit shall have as a minimum the following functional illuminated indicators:

a. AC power ON
b. RF power ON
c. Channel selected

3.2.6 Antenna. The R/T unit shall be designed to operate from a single AS3015/UR antenna.

3.2.7 Handset. The radio set shall be designed to operate with an H169/U handset. / Holder for the H169 shall be provided, along with brackets to attach holder to the control unit.

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Table II. Temperature Range

Equipment	Allowable degradation	Reference paragraph
Transmitter		
Maximum power output	1 dB	3.2.2.1
FM Hum and noise	6 dB	3.2.2.5
Carrier frequency stability	within limits of 3.2.2.7	3.2.2.7
Receiver		
Usable sensitivity	3 dB	3.2.3.1
Audio squelch sensitivity	4 dB	3.2.3.3
Modulation acceptance bandwidth	10 %	3.2.3.4
Adjacent channel selectivity	8 dB	3.2.3.5
Audio power output	1 dB all outputs	3.2.3.8
Hum and noise ratio	8 dB	3.2.3.10
Frequency stability	within limits of 3.2.3.14	3.2.3.14

Tests to be conducted in accordance with 4.4.1.4.

3.3 Reliability.

3.3.1 Reliability. The specified mean-time-between-failure (MTBF) (θ_o as defined in MIL-STD-781) of the radio set shall be 5400 hours.

3.4 Maintainability.

3.4.1 Qualitative maintainability requirements. The equipments shall be constructed to provide ease of maintenance, accessibility, and replacement of all modules and parts.

3.4.2 Maintainability requirements. The equipment shall have an Equipment Repair Time (ERT) of 15 minutes or less.

3.5 Assembly and subassembly interchangeability. Mechanical and electrical interchangeability shall exist between identically numbered assemblies and subassemblies (see 3.5.1) of the radio sets. Provisions shall be made for in design tolerance to accommodate various solid-state components, such as microcircuits and other parts, without departure from the specified performance.

3.5.1 Assemblies and subassemblies. The assemblies and subassemblies listed herein shall be interchangeable between radio sets and with the assemblies and subassemblies of the GFE.

Interchangeable assemblies and subassemblies

A. Receiver-transmitter, radio, RT-1155/URC-80(V)
   1. Power amplifier (A1)
   2. TCXO/Multiplier (A2)
   3. Mixer/Variable divider (A3)
   4. VCO/Modulator (A4)
   5. Audio squelch (A5)
   6. Receiver (A6)
   7. Power supply (A7)
   8. Wired chassis (A8)
   9. Receiver (A9)
   10. Injection oscillator (A10)

B. Control, receiver-transmitter, C-8980/URC-80(V)

C. Loudspeaker, LS-609/UR

D. Antenna, AS-3015/[illegible]

E. Cables

No degradation in performance limits shall be permitted as a result of interchangeability requirements.

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3.6 Vibration. The equipment shall meet the vibration requirements of MIL-STD-167 for Type I lightweight equipment.

3.7 Equipment conditioning. All deliverable equipment shall be subjected to an environmental burn-in (see 4.7) to assure stabilization of required equipment characteristics and elimination of initial failures due to random defective components or inadequate assembly processes. As a minimum, the critical parameters in Table III shall be measured at least once daily and just prior to completion of the equipment conditioning tests to verify proper operation of the equipment. (see 4.4.1.2)

Table III. Minimum Parameter Measurements

Parameter	Reference paragraph	Maximum allowable degradation
RF power output	3.2.2.1	1 dB
Carrier frequency stability	3.2.2.7	within the limits of 3.2.2.7
Usable sensitivity	3.2.3.1	3 dB
Audio power output	3.2.3.8	1 dB @ 5% distortion

No permanent performance degradation should occur as a result of equipment conditioning.

3.8 First article sample. Prior to beginning production, a sample shall be tested as specified in 4.3.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Government verification. All quality assurance operations performed by the contractor will be subject to Government verification at any time. Verification will include (a) surveillance of the operations to determine that practices, methods, and procedures are being properly applied, (b) Government product inspection to measure quality of product to be offered for acceptance and (c) Government product inspection of delivered products to assure compliance with all inspection requirements of this specification. Failure of the contractor to promptly correct deficiencies discovered by him or of which he is notified shall be cause for suspension of acceptance until corrective action has been made or until conformance of product to prescribed criteria has been demonstrated.

4.1.2 Quality assurance terms and definitions. Quality assurance terms used in this specification shall be as defined in MIL-STD-109.

4.2 Classification of inspection. The method of examination and testing shall be classified as follows:

(a) First Article Inspection (see 4.3)
(b) Quality Conformance Inspection (see 4.4)
    (1) Production Inspection (see 4.4.1)
    (2) Production Control Inspection (see 4.4.3)
    (3) Periodic Inspection (see 4.4.4)
    (4) Inspection of Preparation for Delivery (see 4.4.5)
(c) Reliability Testing (see 4.5)
(d) Maintainability Demonstration (see 4.6)

4.3 Unless otherwise specified (see 6.1.1), one unit shall be required for first article inspection. First article inspection shall consist of all examination and testing specified in production inspection (see 4.4.1), production control inspection (see 4.4.3), periodic inspection (see 4.4.4), and the additional first article inspections in Table IV.

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Table IV. Additional First Article Inspections

Inspection	Test Channel	Reference Paragraph	Test Paragraph
Vibration	16	3.6	4.5.7

4.4 Quality conformance inspection.

4.4.1 Production inspection. Production inspection shall be made on every equipment offered for delivery. The inspection shall comprise such examination and testing as will prove the workmanship and reveal the omissions and errors of the production process such as functional and performance tests at a limited number of points, tests which detect deviations from design, tests of adjustment, and tests which detect hidden defects of material. Production inspection shall include the requirements of table V and shall be performed after the equipment conditioning specified in 4.4.1.2.

Table V. Production Inspection

Inspection	Channels		Reference Paragraphs	Test Paragraphs
General				
Surface examination				4.4.1.1
Operating test	16			4.4.1.3
Transmitter				
Frequency accuracy	1,16,88		3.2.2.7	
* R.F. power output (1 watt and 25 watt)	1,16,88		3.2.2.1	
Audio frequency harmonic distortion	16		3.2.2.3	4.5.5
FM hum and noise level	16		3.2.2.5	
Modulation limiting	1,16,88		3.2.2.6	
Modulation and noise spectrum	16		3.2.2.10	
Receiver and guard receiver	RECV'R	GRD.RECV'R		
Usable sensitivity	1,16,88	16	3.2.3.1	
Quieting sensitivity	1,16,88	16	3.2.3.2	4.5.6
Modulation acceptance bandwidth	16	16	3.2.3.4	
Audio power output	16	16	3.2.3.8	
Hum and noise ratio	16	16	3.2.3.10	

*Does not require 8 hour test

4.4.1.1 Surface examination. Equipment shall be examined for obvious defects in workmanship, assembly, fit, and mechanical safety and marking. The examination shall include verification of materials, parts and finish used in the equipment manufacture.

4.4.1.2 Equipment conditioning. All deliverable equipments shall be subjected to burn-in in accordance with 3.7 prior to production inspection. Burn-in shall be performed with the equipment operating in a 1 minute transmit/1 minute receive duty cycle and at specified maximum transmit and audio power levels.

4.4.1.2.1 Accept criteria. Failures occurring during the first 24 hours of high temperature exposure shall be repaired and the burn-in resumed. In the event of failure during the final 24 hours, the equipment shall be repaired and the burn-in shall be resumed with a minimum of 24 hours of successful operation prior to acceptance.

4.4.1.2.2 Rework. In the event an equipment is reworked following initial successful completion of the burn-in test, and the rework is of a minor nature (such as replacement of a single component or group of components), equipment conditioning need not be reperformed. However, should the rework be major (such as replacement of a complete subassembly with a subassembly which has not previously been burned-in), equipment conditioning shall be performed as specified in 4.4.1.2..

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4.4.1.3 Operating test. The equipment shall be energized and subjected to an operating test to insure qualitatively the proper functioning of the equipment including all operating controls (see 3.2.5) and the accessories (see 3.2.6, 3.2.7). This test shall include actual voice communications between two equipments.

4.4.1.4 Temperature test. The parameters shall initially be measured at room temperature after equipment has been stabilized. The equipment shall then be made inoperative and lowered in temperature to 0°C for a 2 hour soak period after which it shall be turned on. Receiver measurements shall be made first followed by the transmitter measurements. All measurements shall be completed within 1 hour. Following completion of the low temperature measures, the temperature shall be raised to + 50°C and allowed to "soak" for 8 hours while being operated at a 50 percent duty cycle of 1 minute transmit and 1 minute receive. Maximum RF power and audio power outputs shall be provided. Measurements shall be made following the soak period. No adjustments of equipment shall be permitted nor shall forced circulation of air be allowed directly on the equipment. (see 3.2.4)

4.4.2 Frequency stability. Frequency stability measurements shall be made by measuring the receiver operating frequency while maintaining the last intermediate frequency (IF) constant.

4.4.3 Production control inspection. Production control inspection, including sampling, shall conform to Table VI below and to the inspection procedures of MIL-STD-105 using the special inspection levels. The inspection level shall be S-3 for normal, tightened, and reduced inspection. Production control inspection shall be performed on equipments that have passed production inspection. The equipment shall satisfactorily meet the requirements of production control inspection prior to release for shipment.

Table VI. Production Control Inspection

Inspection	Test channels	Reference paragraph	AQL %	Test paragraph
Transmitter				
Frequency accuracy	All 55	3.2.2.7	6.5	4.5.5
Audio frequency response		3.2.2.4		4.5.5
Receiver				
Adjacent channel selectivity and desensitization	1, 88	3.2.3.5	6.5	4.5.6
Intermodulation spurious attenuation	16	3.2.3.7	6.5	4.5.6
General				
Antenna overload	16	3.1.12	6.5	–
Power source	16	3.2.1	6.5	4.5.5,4.5.6

4.4.3.1 Lots requiring reinspection. If an inspection lot requires reinspection, the supplier may either withdraw the lot from further inspection and submission or rework a lot to (a) correct the defective units (b) reinspect the lot using tightened inspection. Lots requiring reinspection shall be kept separate from other lots and shall not lose their identity.

4.4.4 Periodic inspection. Periodic inspection shall be as listed in Table VII and shall be performed on units which have been subjected to and passed production inspection.

4.4.4.1 Sampling for periodic inspection. One equipment shall be selected from each successive 50 equipments produced or fraction thereof. The first sample equipment shall be selected from the first months production.

4.4.4.2 Nonconforming periodic sample units. If a sample unit fails the inspection specified in 4.4.3, the contractor shall immediately investigate the cause of failure and shall report to the Quality Assurance Representative the results thereof and details of the corrective action taken to correct units of product which were manufactured under the same conditions, with the same materials and processes. If the Quality Assurance Representative does not consider that the corrective action will enable the product to meet specified requirements, or if the contractor

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cannot determine the cause of failure, the matter shall be referred to the contracting officer (see 6.2).

Table VII. Periodic Inspection

Inspection	Test channels	Reference paragraph	Test paragraph
Transmitter			
RF power output (20-25W)	1,16,88	3.2.2.1	4.5.5
Conducted spurious emissions	1, 88	3.2.2.2.	4.5.5
Humidity	16	3.2.2.8	4.5.5
Receiver and guard receiver			
Spurious response attenuation	1,88	3.2.3.6	4.5.6
Humidity	16	3.2.3.13	4.5.6
General			
Interchangeability	1,16,88	3.5	4.5.4
Temperature range	16	3.2.4	4.4.1.4

4.4.4.3 Reinspection of conforming periodic sample units. Unless otherwise specified in the contract, sample units which have been subjected to, and passed, periodic tests may be accepted on the contract provided they are resubjected to, and pass, production inspection after repair of all damage.

4.4.5 Inspection of preparation for delivery. Inspection shall be conducted to insure conformance with the requirements of Section 5 of this specification.

4.5 Reliability testing. Reliability testing shall be performed in accordance with MIL-STD-781 on at least three units from the first months production. Testing shall be performed to test level C with accept/reject criteria in accordance with test plan VI. On-off cycling and input voltage cycling are required.

4.5.1 Relevant failure. The final decision as to the relevancy or non-relevancy of a failure shall be made by the procuring activity using the "Failure Categories" paragraph of MIL-STD-781 as a guide: Any operating discrepancy or degradation that requires an unscheduled adjustment or calibration to be made, after initial satisfactory operation of the equipment under test, shall be defined as relevant failure.

4.5.2 Rejected equipment. When equipment has been rejected as a result of the reliability test, the whole lot from which it was drawn shall be reworked as necessary, reinspected, and another random sample drawn for another reliability test. In the event that the cause of failure is equipment manufacturer design or part manufacturer designs, corrective action shall be performed on all equipment delivered under the contract.

4.5.3 Disposition of tested equipment. Upon successful completion of reliability testing, tested equipment shall be returned to the production area for re-inspection and re-conditioning as required and shall be subjected to the production tests of paragraph 4.4.1 to assure that it meets the requirements of this specification. After reconditioning, re-inspection, and retesting, the equipment may be delivered as part of the contract quantity provided it meets all terms and conditions of the criteria for acceptance.

4.5.4 Examinations and tests. The contractor shall select tests from this specification, exclusive of temperature, humidity, and vibration to verify the interchangeability requirements. Tests and examinations shall be selected on the basis of being meaningful for verification of interchangeability to the particular assembly or subassembly being interchanged. Where mechanical criteria are not specified, the contractor shall provide sufficient detail to his test methods to verify the mechanical aspects of interchangeability.

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4.5.5 The transmitter shall be tested in accordance with the method of measurement paragraphs of EIA Standard RS-152 except for temperature tests.

4.5.6 The receivers shall be tested in accordance with the method of measurement paragraphs of EIA Standard RS-204, except for temperature test.

4.5.7 Vibration. The radio set, less antenna, shall be vibrated in accordance with MIL-STD-167 Type I without shock mounts.

4.6 Maintainability demonstration. The Equipment Repair Time (ERT) of 3.4.2 shall be demonstrated in accordance with method 4 of MIL-STD-471.

4.7 Burn-in. The burn-in shall be 48 operating hours at 50 ± 5°C with the last 24 hours failure-free. In addition, each equipment shall be operated for a minimum of 8 hours while subjected to 1.0g ± 10 percent vibration at a non-resonant frequency between 20 and 60 Hz. Vibration may be performed simultaneously with or prior to burn-in (see 3.7, 4.4.1.2).

4.8 Documentation. Reports, test procedures, recorded data and corrective action performed by the contractor shall be available to the Government for examination.

5. PREPARATION FOR DELIVERY

(The preparation for delivery requirements specified herein apply only for direct Government procurements. Preparation for delivery requirements of referenced documents listed in Section 2 do not apply unless specifically stated in the contract or order. Preparation for delivery requirements for products procured by contractors shall be specified in the individual order.)

5.1 Preservation, packaging, packing and marking. Unless otherwise specified, preparation for delivery shall be in accordance with the applicable levels of preservation, packaging, packing and marking specified in MIL-E-17555 (see 6.1.1) or MIL-STD-794, as applicable.

6. NOTES

6.1 Intended use. The URC-80(V)5 radio set is intended for shipboard application to satisfy communications requirements in the maritime service in both national and international waterways.

6.1.1 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification.
b. Marking, levels of preservation, and packaging and packing required (see 5.1).
c. Number of first article samples to be submitted if other than specified in 4.3.

6.2 Periodic inspection. Approval to ship may be withheld at the discretion of the Government, pending the decision from the contracting officer on the adequacy of corrective action.

6.2.1 Initial shipments. Initial shipment of end items will be contingent upon successful completion of the periodic inspection and reliability testing.

6.3 Definitions:

a. Simplex mode - a simplex mode exists when the receiver and transmitter of a receiver-transmitter unit operate on the same frequency (f1/f1) but not simultaneously.

b. Half duplex mode - A half duplex mode exists when the receiver and transmitter of a receiver-transmitter unit operates on separate receive and transmit frequencies (f1/f2) but not simultaneously.

c. Failure - Any malfunction or parameter deviation which prevents the radio set from performing within the limits specified herein.

d. GFE description - The GFE is an AN/URC-80(V)5 radio set procured under another contract.

6.4 Supercession data. This specification includes the requirements of Naval Electronic Systems Command contract specification ELEX-T-134.

Preparing activity
Navy-EC
(Project 5820-N661)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

INSTRUCTIONS: The purpose of this form is to solicit beneficial comments which will help achieve procurement of suitable products at reasonable cost and minimum delay or will otherwise enhance use of the document. DoD contractors, government activities, or manufacturers/vendors who are prospective suppliers of the product are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity.

DOCUMENT IDENTIFIER AND TITLE

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

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